

# Commerce XI Maths App

## Exercise 1.2

Detailed Answers with Explanations and Steps

Mathematics & Statistics, Part 2, Unit 1 - Partition Values | Total questions: 18

### Question 1 - Enter answer · PG-15 · EX-Q1(i)

For data : 79, 82, 36, 38, 51, 72, 68, 70, 64, 63

Calculate  $D_6$ .

**Correct answer: 69.2**

Explanation and steps

Use  $i(n+1)/10$  or  $i(n+1)/100$  for raw data, and the corresponding grouped-data interpolation formula when classes are given. The result is 69.2.

### Question 2 - Enter answer · PG-15 · EX-Q1(ii)

For data : 79, 82, 36, 38, 51, 72, 68, 70, 64, 63

Calculate  $P_{85}$ .

**Correct answer: 80.05**

Explanation and steps

Use  $i(n+1)/10$  or  $i(n+1)/100$  for raw data, and the corresponding grouped-data interpolation formula when classes are given. The result is 80.05.

### Question 3 - Enter answer · PG-15 · EX-Q2(i)

The daily wages (in ₹) of 15 labours are

230, 400, 350, 200, 250, 380, 210, 225, 375, 180, 375, 450, 300, 350, 250

Calculate  $D_8$ .

**Correct answer: 379**

Explanation and steps

Use  $i(n+1)/10$  or  $i(n+1)/100$  for raw data, and the corresponding grouped-data interpolation formula when classes are given. The result is 379.

### Question 4 - Enter answer · PG-15 · EX-Q2(ii)

The daily wages (in ₹) of 15 labours are

230, 400, 350, 200, 250, 380, 210, 225, 375, 180, 375, 450, 300, 350, 250

Calculate  $P_{90}$ .

**Correct answer: 420**

Explanation and steps

Use  $i(n+1)/10$  or  $i(n+1)/100$  for raw data, and the corresponding grouped-data interpolation formula when classes are given. The result is 420.

### Question 5 - Enter answer · PG-15 · EX-Q3(i)

Calculate 2nd decile for the following :

| x | 80 | 100 | 120 | 145 | 200 | 280 | 310 | 380 | 400 | 410 |
|---|----|-----|-----|-----|-----|-----|-----|-----|-----|-----|
| f | 15 | 18  | 25  | 27  | 40  | 25  | 19  | 16  | 8   | 7   |

**Correct answer: 120**

Explanation and steps

Use  $i(n+1)/10$  or  $i(n+1)/100$  for raw data, and the corresponding grouped-data interpolation formula when classes are given. The result is 120.

### Question 6 - Enter answer · PG-15 · EX-Q3(ii)

Calculate 65th percentile for the following :

| x | 80 | 100 | 120 | 145 | 200 | 280 | 310 | 380 | 400 | 410 |
|---|----|-----|-----|-----|-----|-----|-----|-----|-----|-----|
| f | 15 | 18  | 25  | 27  | 40  | 25  | 19  | 16  | 8   | 7   |

**Correct answer: 280**

Explanation and steps

Use  $i(n+1)/10$  or  $i(n+1)/100$  for raw data, and the corresponding grouped-data interpolation formula when classes are given. The result is 280.

### Question 7 - Enter answer · PG-15 · EX-Q4(i)

From the following data calculate the rent of 15th house.

| House Rent (in ₹) | 11000 | 12000 | 13000 | 14000 | 15000 | 16000 | 17000 | 18000 |
|-------------------|-------|-------|-------|-------|-------|-------|-------|-------|
| No. of houses     | 25    | 17    | 13    | 14    | 15    | 8     | 6     | 2     |

**Correct answer: 11000**

Explanation and steps

Use  $i(n+1)/10$  or  $i(n+1)/100$  for raw data, and the corresponding grouped-data interpolation formula when classes are given. The result is 11000.

### Question 8 - Enter answer · PG-15 · EX-Q4(ii)

From the following data calculate the rent of 65th house.

| House Rent (in ₹) | 11000 | 12000 | 13000 | 14000 | 15000 | 16000 | 17000 | 18000 |
|-------------------|-------|-------|-------|-------|-------|-------|-------|-------|
| No. of houses     | 25    | 17    | 13    | 14    | 15    | 8     | 6     | 2     |

**Correct answer: 14000**

Explanation and steps

Use  $i(n+1)/10$  or  $i(n+1)/100$  for raw data, and the corresponding grouped-data interpolation formula when classes are given. The result is 14000.

### Question 9 - Enter answer · PG-15 · EX-Q4(iii)

From the following data calculate the rent of 91st house.

| House Rent (in ₹) | 11000 | 12000 | 13000 | 14000 | 15000 | 16000 | 17000 | 18000 |
|-------------------|-------|-------|-------|-------|-------|-------|-------|-------|
| No. of houses     | 25    | 17    | 13    | 14    | 15    | 8     | 6     | 2     |

**Correct answer: 16000**

Explanation and steps

Use  $i(n+1)/10$  or  $i(n+1)/100$  for raw data, and the corresponding grouped-data interpolation formula when classes are given. The result is 16000.

### Question 10 - Enter answer · PG-15 · EX-Q5(i)

The following frequency distribution shows the weight of students in a class.

| Weight (in kg)     | 40 | 45 | 50 | 55 | 60 | 65 |
|--------------------|----|----|----|----|----|----|
| Number of Students | 15 | 40 | 29 | 21 | 10 | 5  |

Find the percentage of students whose weight is more than 50 kg.

**Correct answer: 30**

Explanation and steps

Use  $i(n+1)/10$  or  $i(n+1)/100$  for raw data, and the corresponding grouped-data interpolation formula when classes are given. The result is 30.

### Question 11 - Enter answer · PG-15 · EX-Q5(ii)

The following frequency distribution shows the weight of students in a class.

| Weight (in kg)     | 40 | 45 | 50 | 55 | 60 | 65 |
|--------------------|----|----|----|----|----|----|
| Number of Students | 15 | 40 | 29 | 21 | 10 | 5  |

If the weight column provided is of mid values then,  
find the percentage of students whose weight is more than 50 kg.

**Correct answer: 42**

Explanation and steps

Use  $i(n+1)/10$  or  $i(n+1)/100$  for raw data, and the corresponding grouped-data interpolation formula when classes are given. The result is 42.

### Question 12 - Enter answer · PG-15 · EX-Q6(i)

Calculate  $D_4$  from the following data:

| Mid Value | 2.5 | 7.5 | 12.5 | 17.5 | 22.5 | Total |
|-----------|-----|-----|------|------|------|-------|
| Frequency | 7   | 18  | 25   | 30   | 20   | 100   |

**Correct answer: 13**

Explanation and steps

Use  $i(n+1)/10$  or  $i(n+1)/100$  for raw data, and the corresponding grouped-data interpolation formula when classes are given. The result is 13.

### Question 13 - Enter answer · PG-15 · EX-Q6(ii)

Calculate  $P_{48}$  from the following data:

| Mid Value | 2.5 | 7.5 | 12.5 | 17.5 | 22.5 | Total |
|-----------|-----|-----|------|------|------|-------|
| Frequency | 7   | 18  | 25   | 30   | 20   | 100   |

**Correct answer: 14.6**

Explanation and steps

Use  $i(n+1)/10$  or  $i(n+1)/100$  for raw data, and the corresponding grouped-data interpolation formula when classes are given. The result is 14.6.

### Question 14 - Enter answer · PG-15 · EX-Q7(i)

Calculate  $D_9$  of the following distribution.

| Length (in Inches) | 0-20 | 20-40 | 40-60 | 60-80 | 80-100 | 100-120 |
|--------------------|------|-------|-------|-------|--------|---------|
| No. of units       | 1    | 14    | 35    | 85    | 90     | 15      |

**Correct answer: 98**

Explanation and steps

Use  $i(n+1)/10$  or  $i(n+1)/100$  for raw data, and the corresponding grouped-data interpolation formula when classes are given. The result is 98.

### Question 15 - Enter answer · PG-15 · EX-Q7(ii)

Calculate  $P_{20}$  of the following distribution.

| Length (in Inches) | 0-20 | 20-40 | 40-60 | 60-80 | 80-100 | 100-120 |
|--------------------|------|-------|-------|-------|--------|---------|
| No. of units       | 1    | 14    | 35    | 85    | 90     | 15      |

**Correct answer: 58.86**

Explanation and steps

Use  $i(n+1)/10$  or  $i(n+1)/100$  for raw data, and the corresponding grouped-data interpolation formula when classes are given. The result is 58.86.

### Question 16 - MCQ · PG-16 · EX-Q8

Weekly wages for group of 100 persons are given below :

| Weekly wages (in Rs.) | 0-500 | 500-1000 | 1000-1500 | 1500-2000 | 2000-2500 |
|-----------------------|-------|----------|-----------|-----------|-----------|
| No. of persons        | 7     | a        | 25        | 30        | b         |

$D_3$  for this group is Rs. 1100. Calculate the missing frequencies a, b.

Options

- A. (18, 20)
- B. (20, 18)
- C. (17, 21)
- D. (22, 16)

**Correct answer: (18, 20)**

Explanation and steps

Use  $i(n+1)/10$  or  $i(n+1)/100$  for raw data, and the corresponding grouped-data interpolation formula when classes are given. The result is (18, 20).

### Question 17 - MCQ · PG-16 · EX-Q9

The weekly profit (in rupees) of 100 shops are distributed as follows :

| Profit per shop | 0-1000 | 1000-2000 | 2000-3000 | 3000-4000 | 4000-5000 | 5000-6000 | 6000-7000 |
|-----------------|--------|-----------|-----------|-----------|-----------|-----------|-----------|
| No. of Shops    | 10     | 16        | 26        | 20        | 20        | 5         | 3         |

Find the limits of the profit of middle 60% of the shops.

Options

- A.  $(P_{20}, P_{80}) = (1625, 4400)$
- B.  $(P_{20}, P_{80}) = (1500, 4500)$
- C.  $(P_{30}, P_{70}) = (1625, 4400)$
- D.  $(P_{20}, P_{80}) = (1600, 4600)$

**Correct answer:  $(P_{20}, P_{80}) = (1625, 4400)$**

Explanation and steps

Use  $i(n+1)/10$  or  $i(n+1)/100$  for raw data, and the corresponding grouped-data interpolation formula when classes are given. The result is  $(P_{20}, P_{80}) = (1625, 4400)$ .

### Question 18 - Enter answer · PG-16 · EX-Q10

In a particular factory, workers produce various types of output units.  
The following distribution was obtained

| Output units Produced | 70-74 | 75-79 | 80-84 | 85-89 | 90-94 | 95-99 | 100-104 |
|-----------------------|-------|-------|-------|-------|-------|-------|---------|
| No. of workers        | 40    | 45    | 50    | 60    | 70    | 80    | 100     |

Find the percentage of workers who have produced less than 82 output units.

**Correct answer: 24.72**

Explanation and steps

Use  $i(n+1)/10$  or  $i(n+1)/100$  for raw data, and the corresponding grouped-data interpolation formula when classes are given. The result is 24.72.